Folake Oluewu - CS 424 Assignment 2

Task 1

My project explores how activity differs across three coffee shops: Afro Joe's Coffee & Tea, Cafe 53, and Philz Coffee. I collected 54 observations, with 18 observations per coffee shop, recording activity at different times and on different dates.

My main research questions are:

1. How does the number of people in each coffee shop vary by observation time?
2. What is the average number of people present at each coffee shop?
3. Which coffee shop has the most people working or studying?
4. How do the number of people and available seats compare?
5. How does the number of people waiting in line differ between coffee shops?

One limitation is that observations were collected at different times and on different dates, rather than continuously. Factors such as weather, special events, and exact seating capacity were not consistently recorded.

Task 2

I organized my observations into a CSV file and used Python with pandas to prepare and inspect the data. I used a cleaning script to check the dataset and create a processed CSV file.

My dataset contains 54 observations, with 18 observations for each coffee shop. The checks reported no missing values and no duplicate rows.

I kept the original dataset and the processed dataset separate so that the original data would remain available. My Python scripts use the processed CSV file for the analysis and visualizations.

Task 3

I used Python and pandas to calculate average values for each coffee shop and examine how the number of people varied by observation time.

Some findings from my exploration were:

- **Average people counted:** Afro Joe's had 11.44, Philz Coffee had 11.28, and Cafe 53 had 5.28.
- **Average people working:** Philz Coffee had 6.56, Afro Joe's had 6.44, and Cafe 53 had 1.06.

- **Average people in line:** Philz Coffee had 2.94, Afro Joe's had 2.89, and Cafe 53 had 1.44.
- **Average available seats:** Afro Joe's had 8.83, Philz Coffee had 7.72, and Cafe 53 had 5.28.

These results suggest that Afro Joe's and Philz had more people counted, more people working, and longer lines on average than Cafe 53 in my observations.

However, these are descriptive averages from my sample, not proof that the same patterns occur every day. The observations were collected at different times and dates.

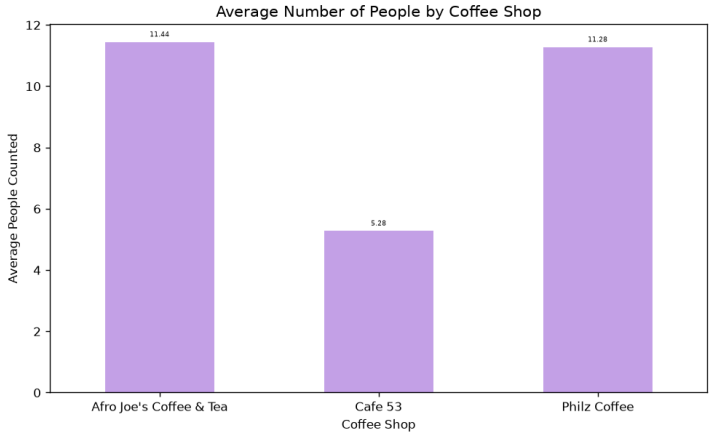
Task 4

I created three visualizations to explore my data: a bar chart, a line chart, and a heatmap. I used the same dataset and focused on comparing the number of people counted.

Bar Chart

The bar chart compares the average number of people counted at each coffee shop. I added numerical labels above the bars so viewers can see the exact averages without estimating them from the axis. I also adjusted the colors to make the chart easier to distinguish visually.

Screenshot:

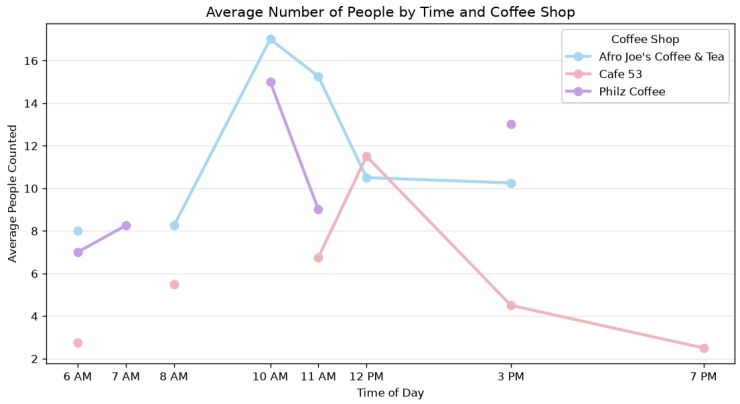


Line Chart

The line chart compares average people counts across observation times for each coffee shop. I used different colors and markers to distinguish the shops and make the recorded values easier to identify.

Because observations were collected at different times and dates, the connecting lines do not represent continuous tracking of the shops.

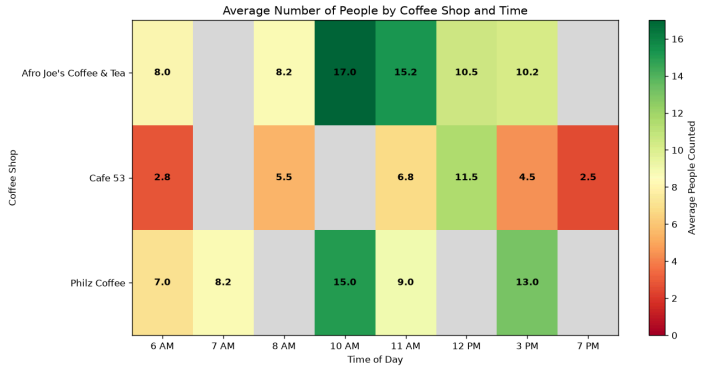
Screenshot:



Heatmap

The heatmap compares average people counts across coffee shops and observation times. The color scale helps viewers identify higher and lower values, while numerical labels show the actual averages. I changed the color scale to red, yellow, and green to make differences more visible. Gray cells represent combinations for which no observations were collected.

Screenshot:



Task 5

I explored three different ways to visualize the same data.

1. Bar chart: This is useful for comparing average people counts between coffee shops. It makes it easy to identify which shop had the highest or lowest average, but it does not show how counts compare across observation times.

2. Line chart: This makes it easier to compare recorded averages across times for different coffee shops. However, connecting the points may suggest a continuous trend even though the observations were collected separately.

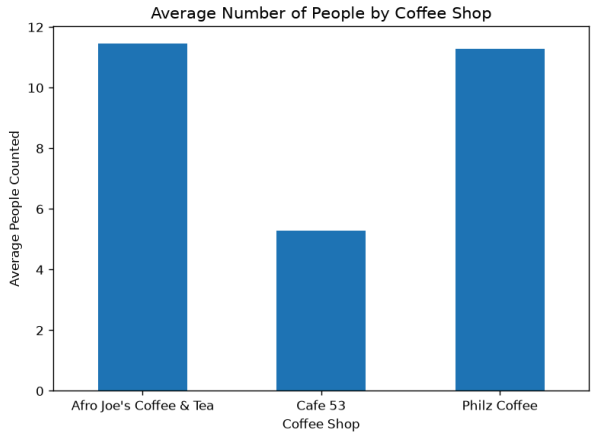
3. Heatmap: This shows the relationship between coffee shop and observation time in one view. The colors make higher and lower averages easier to spot, although viewers may need to refer to the numerical labels for precise comparisons.

Each visualization emphasizes a different aspect of the data. Comparing them helped me understand that the best chart depends on what information I want viewers to notice.

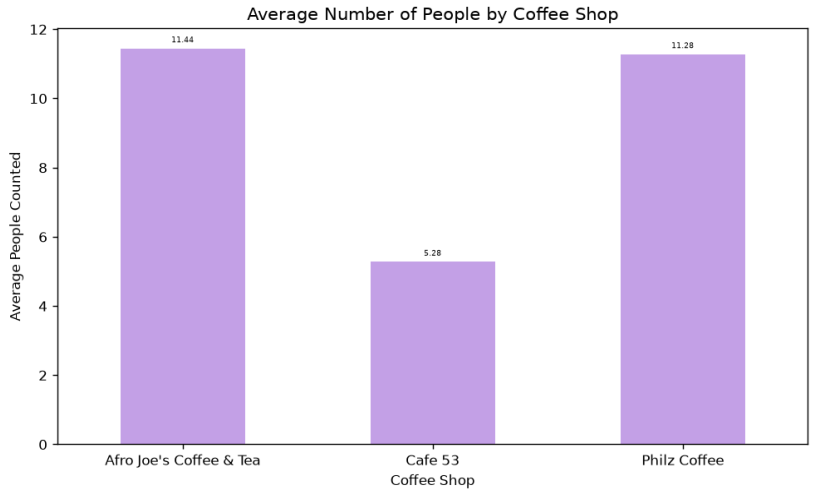
Task 6

Iteration 1: Bar Chart

Before screenshot:



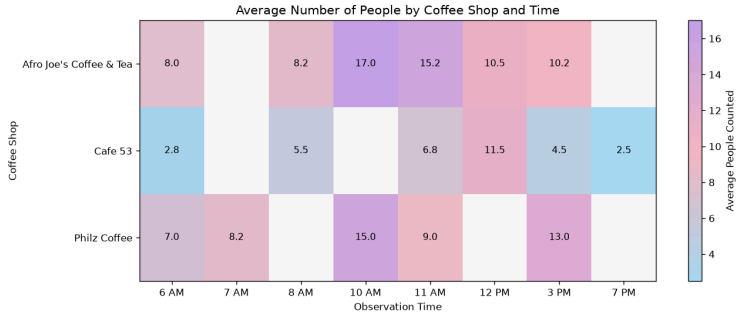
After screenshot:



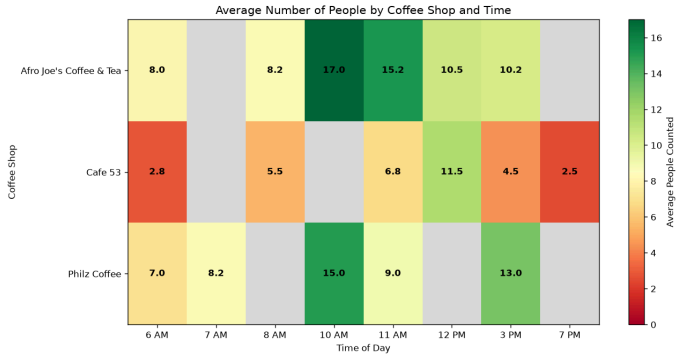
I revised the bar chart by adding numerical labels above the bars and adjusting the colors. These changes make the exact averages easier to read and improve the visual distinction between the coffee shops.

Iteration 2: Heatmap

Before screenshot:



After screenshot:



I changed the heatmap from a pastel color scheme to a more contrasting red, yellow, and green gradient. I also made the numerical labels black so they remain visible against the different background colors. This makes differences between the data points easier to notice while keeping the exact averages available.

These iterations helped me understand how color, labels, and visual contrast affect how easily people can interpret a visualization.

Task 7

For Assignment 3, I would like to continue developing the heatmap because it allows viewers to compare coffee shops and observation times in one view. The contrasting color scale makes patterns easier to notice, and the numerical labels provide more precise information.

I would like to explore ways to make the visualization more interactive, such as allowing viewers to select different measurements, including people working, people in line, and available seats. This would make the visualization more useful for exploring the other research questions in my project.

I would also like to improve how the visualization communicates missing observations and the limitations of the dataset, so viewers do not mistake unrecorded values for zero or interpret the results as continuous tracking.